

Econometrics Research Experiment: Hawaii GDP

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Introduction

The purpose of this experiment is to assess the determinants of Hawaii's real GDP, specifically testing the "common wisdom" that the island's economy is driven primarily by Tourism and Military spending. I also aim to quantify the non-linear effects of Oil Prices (given Hawaii's unique petroleum dependence) and the structural break caused by the COVID-19 pandemic.

Data Description and Sources: This analysis utilizes quarterly data from Q1 2013 through Q4 2024 (n=48). Below is a description of the variables, their specific sources, and the economic rationale for their inclusion.

1. Dependent Variable: Real Gross Domestic Product (GDP)

- Definition: All Industry Total in the State of Hawaii (Series HIRQGSP).
- Source: U.S. Bureau of Economic Analysis via FRED (2025, October 26).
- Rationale: Based on economic theory, real (inflation-adjusted) dollars better reflect the purchasing power of Hawaii's consumers relative to the price of goods. The data is seasonally adjusted to give a cleaner signal of underlying trends by removing seasonal patterns, such as summer and winter tourist cycles.

2. Explanatory Variable: Global Price of Brent Crude

- Hypothesis: Oil prices have a non-linear impact on Hawaii's GDP ($H_0 \geq 0$).
- Sources: International Monetary Fund via FRED (2025); Federal Reserve Bank of St. Louis (2020); Hofschneider (2022). Description: A quarterly time-series of the global Brent Crude spot price.
- Rationale: Historically, Hawaii imported approximately 30% of its oil from Russian sources (Hofschneider, 2022). Although the supply mix shifted after 2022, Brent better captures Hawaii's reliance on imported petroleum than the domestic U.S. West Texas Intermediate (WTI). Brent tracks worldwide oil price movements and typically trades at a modest premium to WTI (FRED, 2020). This premium is justified for Hawaii considering the additional transportation cost of shipping crude to the isolated islands.

3. Explanatory Variable: Total Real Visitor Expenditures

- Hypothesis: Visitor Expenditure has a positive multiplier effect on Hawaii's GDP ($H_0 \geq 0$).
- Definition: A quarterly time-series evaluating real total visitor expenditures in the State of Hawaii, seasonally adjusted to smooth cycles.
- Source: University of Hawai'i Economic Research Organization (UHERO) (2025). - Rationale: Tourism is the engine of the state economy. UHERO forecasts consistently link recessions in Hawaii to declines in visitor arrival numbers. Data Disclosure: Quarters 2, 3, and 4 of 2020 are missing level and change values due to COVID-19 reporting disruptions. I have imputed these missing entries as 0.01 to facilitate log-log analysis, which cannot process zeros or negative values. Additionally, 2021 data lacks year-over-year change metrics because the 2020 baseline is unavailable.

4. Explanatory Variable: Federal Obligations (Defense & Non-Defense)

- Hypothesis: Federal spending has a positive multiplier effect on Hawaii's GDP ($H_0 \geq 0$).
- Sources: USAspending.gov (2025); Hawaii Defense Economy (2023); Leib/ABC News (2025).
- Rationale: The defense sector is the second-largest component of Hawaii's economy, accounting for approximately 8.9% of GDP or roughly \$6,081 per resident. Data Disclosure: To adjust for inflation, I utilize the Consumer Price Index (CPI) for All Urban Consumers in Urban Hawaii. Although the CPI is chained to a 1982 base (while Real GDP is chained to 2017), the regression results remain valid as logarithmic transformation removes constant scaling. Because this CPI data is only available semi-annually, I applied a 2-quarter moving average to interpolate quarterly values.

5. Control & Instrumental Variables To ensure the robustness of the model, two additional variables were introduced:

a. COVID-19 Dummy (Binary):

- A dummy variable set to "1" for the period Q1 2020 through Q2 2021. This captures the exogenous shock of the pandemic, specifically the period covering Hawaii's mandatory quarantine. This period ends as the government of Hawaii began easing Covid restrictions to welcome international visitors in late 2021 (Medina, 2021).
- Source: Medina, E. (2021, November 4). Hawaii Eases Covid Restrictions and Reopens for Tourists. The New York Times.

b. Defense Spending per Personnel (Instrument):

- Sourced from annual reports by the U.S. Department of Defense Office of Local Defense Community Cooperation (OLDCC), this variable measures defense spending intensity relative to personnel count. This is used as an Instrumental Variable (IV) to correct for the potential endogeneity of total federal spending (correcting for reverse causality where the government spends more because the economy is weak).
- Source: OLDCC (2024). Defense Spending by State Fiscal Year 2024.

Summary Statistics

The table below summarizes the key variables used in this analysis (n=48).

```
# 1. Load Data
data <- read.csv("C:/Users/cgrtw/Desktop/Econometrics Class/Datasets/Anthony Zhong - Research Experiment")

# 2. Date Conversion
data$year_quarter <- zoo::as.yearqtr(data$year_quarter, format = "%YQ%q")

# 3. Create Covid Variable (Binary)
data$covid <- ifelse(data$year_quarter >= "2020 Q1" &
                     data$year_quarter <= "2021 Q2", 1, 0)

# 4. Create Logs and Quadratics
data$gdp <- data$hi_gdp_real_2017_millions_usd_sa
data$lgdp <- log(data$gdp)

data$oil <- data$global_brent_bbl_nominal_usd_nsa
data$loil <- log(data$oil)
data$loilsq <- log(data$oil)^2

data$visitor <- data$hi_visitor_exp_real_2024_millions_usd_sa
data$lvisitor <- log(data$visitor)
```

```
data$lvisitorsq <- log(data$visitor)^2

data$federal    <- data$hi_fed_oblig_real_1984_millions_usd_nsa_2qtr_ma
data$lfederal   <- log(data$federal)
data$lfederalsq <- log(data$federal)^2

data$dodIntensity <- data$dod_hi_annual_spending_per_personnel_nominal_usd_4qrt_avg
data$ldodIntensity <- log(data$dodIntensity)
data$ldodIntensitysq <- log(data$dodIntensity)^2
```

```
subset_data <- data %>% select(lgdp, loil, lvisitor, lfederal, ldodIntensity)

stargazer(subset_data, type = "latex", digits = 2,
  title = "Summary Statistics",
  summary.stat = c("n", "mean", "sd", "min", "max"))
```

% Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac at gmail.com % Date and time: Fri, Dec 12, 2025 - 2:12:08 PM

Table 1: Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Max
lgdp	48	11.36	0.05	11.22	11.44
loil	48	4.24	0.32	3.51	4.73
lvisitor	48	7.69	3.21	-4.61	8.63
lfederal	48	7.16	0.29	6.57	7.71
ldodIntensity	48	10.22	0.19	9.85	10.63

Main Analysis

We estimate two models to explain variations in Hawaii's Log GDP.

1. Model 1 (Simple): A baseline regression using Visitor Expenditure.
2. Model 2 (Quadratic): Includes Brent Oil Prices, Total Federal Spending, and Covid controls with quadratic terms.

```
# Model 1: Simple Regression
one.lm <- lm(lgdp ~ lvisitor, data = data)

# Model 2: Multivariate Quadratic Regression
two.lm <- lm(lgdp ~ loil + loilsq +
  lvisitor + lvisitorsq +
  lfederal + lfederalsq +
  covid, data = data)
```

```
stargazer(one.lm,two.lm,type = "latex",omit.table.layout = "n",omit.stat = "ser", column.labels=c("Simple", "Quadratic"))
```

% Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac at gmail.com % Date and time: Fri, Dec 12, 2025 - 2:12:08 PM

Table 2:

	<i>Dependent variable:</i>	
	lgdp	
	Simple Model	Quadratic Model
	(1)	(2)
loil		1.597*** (0.340)
loilsq		-0.192*** (0.040)
lvisitor	0.009*** (0.002)	-0.027** (0.010)
lvisitorsq		0.008*** (0.003)
lfederal		1.692** (0.653)
lfederalsq		-0.116** (0.046)
covid		-0.001 (0.024)
Constant	11.292*** (0.015)	1.541 (2.340)
Observations	48	48
R ²	0.334	0.737
Adjusted R ²	0.320	0.691
F Statistic	23.104*** (df = 1; 46)	15.999*** (df = 7; 40)

Turning Points (Vertex) - To interpret the results for further analysis, I will calculate the vertex of coefficients in the quadratic model.

```
# Vertex for Oil
b1 <- coef(two.lm)["loil"]
b2 <- coef(two.lm)["loilsq"]
oil_vertex <- round(-b1 / (2 * b2),2);

# Vertex for Visitor
b1 <- coef(two.lm)["lvisitor"]
b2 <- coef(two.lm)["lvisitorsq"]
visitor_vertex <- round(-b1 / (2 * b2),2);

# Vertex for Federal
b1 <- coef(two.lm)["lfederal"]
b2 <- coef(two.lm)["lfederalsq"]
federal_vertex <- round(-b1 / (2 * b2),2);

# Create a table with column names and values
my_table <- data.frame(
  oil_vertex,
  visitor_vertex,
  federal_vertex
)

colnames(my_table) <- c("lOil", "lVisitor", "lFederal")
rownames(my_table) <- "vertex value"

my_table

##           lOil lVisitor lFederal
## vertex value 4.16      1.59      7.26
```

Interpretation of Results:

The Quadratic Model significantly improves the correlation (adjusted R-square) compared to the Simple Model.

1. Oil Prices: The model reveals a non-linear relationship. The statistically significant coefficients for loil (positive) and loilsq (negative) suggest a “U-shape” effect where Oil prices positively impact GDP until a certain threshold (calculated vertex approx 4.16), after which they become a drag on the economy. While it looks counter-intuitive, it suggests there may be a positive correlation between global demand and oil prices, up to the point where energy costs become too expensive and negatively affects GDP.
2. Visitor Expenditure: The model also reveals both lvisitor (linear and negative) and lvisitorsq (squared term and positive) are statistically significant at the 5% level. This signifies the model finds visitor expenditure to negatively impact Hawaii’s GDP until a certain point. After this point, visitor expenditure increases GDP. However, the calculated turning point (vertex approx 1.59) is quite low, and could be the quadratic term might be overfitting a relationship that is more effective as linear.
3. Total Federal Spending: The model also reveals both lfederal (linear and positive) and lfederalq (squared term and negative) are statistically significant at the 5% level. However, given that federal spending hiked as a reaction to a decrease in GDP numbers during Covid-19 restrictions, this suggest the variable may be endogenous as it is affect from past errors.

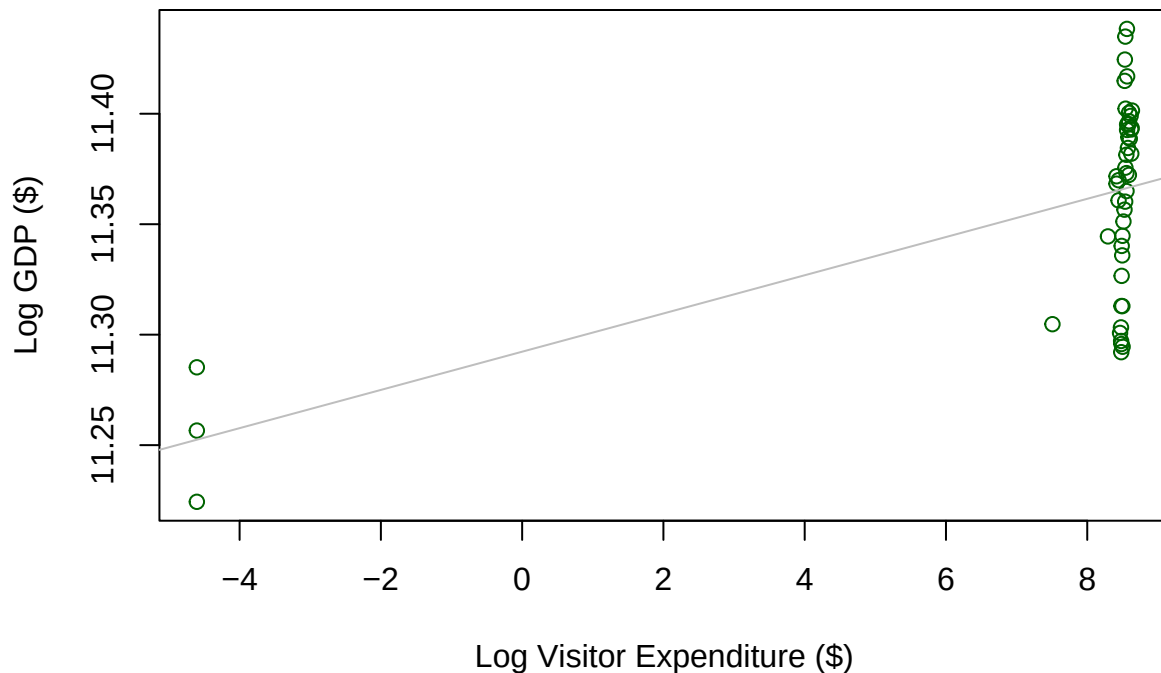
4. Covid-19: The dummy variable is negative but not statistically significant. This is counter-intuitive as Covid did have a large influence in affecting worldwide, but it likely occurs because the `lvisitor` and `lfederal` variable already captures the drop in GDP during the pandemic (Covid travel restriction -> Less visitors -> Less visitor expenditure -> lower GDP) and government fiscal spending that greatly increased up when the covid restrictions happened to stimulate the economy. The dummy variable is, thus, likely picking up residual variance.

Scatter Plot

The following plot illustrates the relationship between the dependent variable (Log GDP) and the main independent variable (Log Visitor Expenditure). The line represents the fitted values from Model 1.

```
# Scatter plot with Regression Line
plot(data$lvisitor, data$lgdp,
     xlab="Log Visitor Expenditure ($)",
     ylab="Log GDP ($)", col="darkgreen")

abline(one.lm, col="gray")
```



Six Assumptions

Assumption 1: Linearity

Definition: The model is linear in parameters and that in the population, the relationship between y and the explanatory variable is linear. $Y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k + u$

Passed: The coefficients are linear, and by including quadratic terms (loilsq, lvisitorsq), this model can still model non-linear curvature while keeping the equation linear in parameters. ($y = \beta_0 + \beta_1 loil + \beta_2 loil^2 + \dots + \beta_7 covid$)

Assumption 2: Random Sampling

Definition: The observations are independent and randomly sampled. $(x_{i1}, x_{i2}, \dots, x_{ik}, y_i) : i = 1, \dots, n$

Violation: My data tracks the same entity (Hawaii) over time. The economy in one quarter is naturally related to the health to the previous quarter (persistence). Because the observations are not independent and identically distributed, the significance errors may be biased. However, this can be fixed in robust standard errors to correct the random sampling's standard error.

```
coeftest(two.lm, vcov = vcovHC(two.lm, type="HC1"))
```

```
##
## t test of coefficients:
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.5406085   2.1239772   0.7253 0.4724651
## loil         1.5974869   0.3852834   4.1463 0.0001707 ***
## loilsq       -0.1918335   0.0452496  -4.2395 0.0001284 ***
## lvisitor     -0.0267814   0.0080424  -3.3300 0.0018746 **
## lvisitorsq    0.0083994   0.0022995   3.6528 0.0007444 ***
## lfederal     1.6924777   0.5911517   2.8630 0.0066490 **
## lfederalsq   -0.1164826   0.0412916  -2.8210 0.0074174 **
## covid       -0.0012451   0.0178784  -0.0696 0.9448267
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Assumption 3: No Perfect Collinearity

Definition: In the sample (and therefore the population), none of the independent variables is constant and there are no exact linear relationship between the independent variables.

Passed: R is able to estimate all coefficients in the model. It would not have done so if there exists an independent variable that has an exact linear relationship of another.

Assumption 4: Zero Conditional Mean (Endogeneity)

Definition: The expected value of the explanatory variables must contain no information about the mean of the unobserved factors. Failing means OLS estimators are biased. $E(u_i | x_{i1}, x_{i2}, \dots, x_{ik}) = 0$

Violation: Although Federal Spending is statistically significant in the Quadratic Model, as seen in the interpretation of model 2, I suspect it suffers from endogeneity.

Federal spending is many times reactionary; the government increases spending because the economy is performing poorly. This means Y (GDP) causes X (Federal Spending) just as much as X causes Y . Even though the coefficient is significant, this two-way relationship biases the magnitude of the estimates.

Solution: In future research, I would correct this using an Instrumental Variable like “Defense Spending per Personnel” (ldodIntensity). Military spending in Hawaii is likely determined by Indo-Pacific region geopolitical threats (exogenous), and not related to Hawaii’s state GDP, making it a valid instrument to isolate the true causal effect of federal funds.

For now, I'll test the assumption that the instrument is relevant.

```
# Regress the endogenous variable (Federal Spending) on the instrument (Defense Spending per Personnel)
```

```
relevance_test <- lm(lfederal ~ ldodIntensity, data = data)
summary(relevance_test)
```

```
##
## Call:
## lm(formula = lfederal ~ ldodIntensity, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.52237 -0.15371 -0.05200  0.08129  0.56051
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.4913     2.1156   0.232  0.81737
## ldodIntensity    0.6526     0.2069   3.154  0.00284 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2629 on 46 degrees of freedom
## (20 observations deleted due to missingness)
## Multiple R-squared:  0.1778, Adjusted R-squared:  0.1599
## F-statistic: 9.947 on 1 and 46 DF,  p-value: 0.002837
```

Result: The p-value for ldodIntensity is less than 0.05. This proves the instrument is relevant.

Assumption 5: Homoskedasticity

Definition: The value of the explanatory variables must contain no information about the variance of the unobserved factors. $Var(u_i | x_{i1}, x_{i2}, \dots, x_{ik}) = \sigma^2$. We want the spread of the residuals to not systematically change with the predicted values.

We test for constant variance of the errors using the Breusch-Pagan test.

```
bptest(two.lm)
```

```
##
## studentized Breusch-Pagan test
##
## data:  two.lm
## BP = 5.2094, df = 7, p-value = 0.6344
```

Violation: The p-value is greater than 0.05, so it fails to reject the null hypothesis of homoskedasticity. Nevertheless, I can correct standard errors using robust adjusted standard errors.

Assumption 6: Normality of Errors

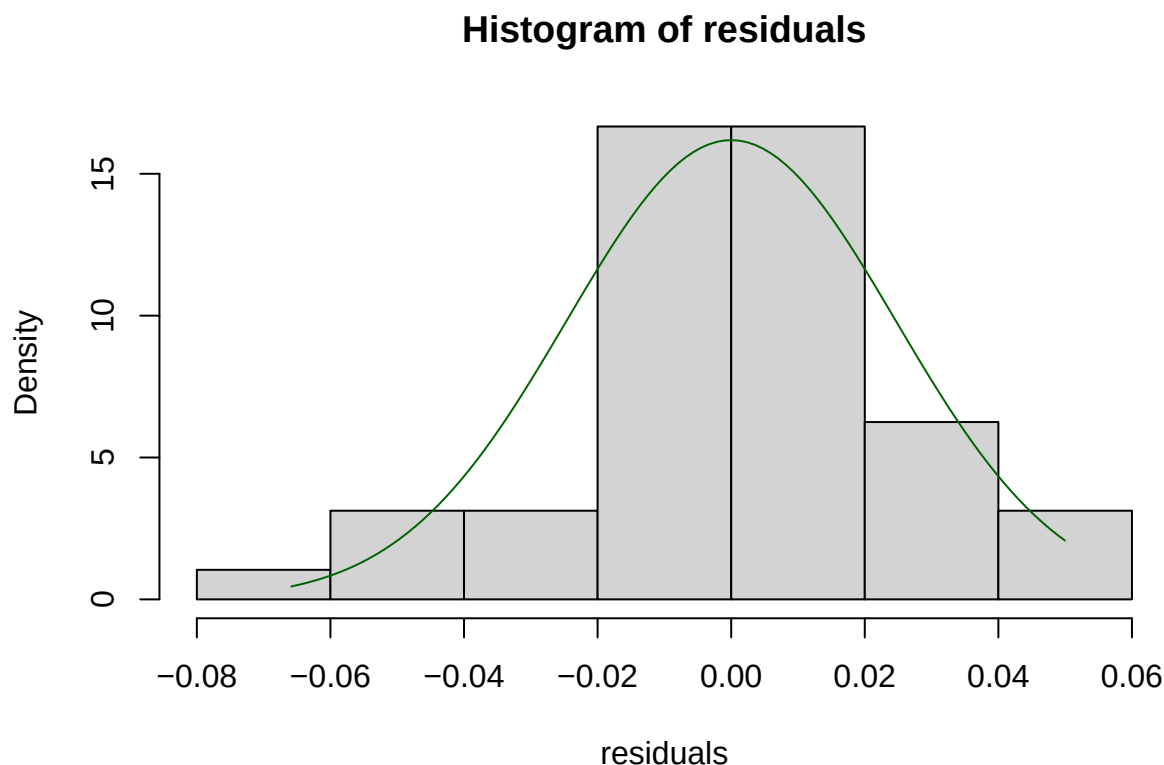
Definition: The residuals of the model are normally distributed independently of the explanatory variables. $u_i \text{ Normal}(0, \sigma^2)$ independently of $x_{i1}, x_{i2}, \dots, x_{ik}$

While the normality assumption is less critical for large sample sizes due to the central limit theorem, it remains important for small observation samples like mines ($n < 100$).

We inspect the residuals to see if they follow a Normal distribution (Bell Curve).

```
residuals <- resid(two.lm)
steps <- (max(residuals)-min(residuals))/100

hist(residuals,freq=F)
lines(seq(min(residuals), max(residuals), by=steps), dnorm(seq(min(residuals), max(residuals), by=steps),
```



Passed: The histogram generally follows the bell curve, suggesting the errors are approximately normally distributed, satisfying the assumption that the residuals of the model are normally distributed.

Conclusion

This analysis confirms that Visitor Expenditure, as common wisdom says, is the primary driver of Hawaii's GDP. The model results indicate:

1. Oil Prices exhibit a statistically significant non-linear impact, showcasing that while energy costs track with economic activity and demand, high prices eventually create a drag on growth.
2. Federal Spending is statistically significant in the quadratic specification. However, this result is likely influenced by endogeneity (reverse causality), where the government spends more in response to economic downturns.

3. Covid-19 was statistically insignificant, likely because the structural break was already captured by the sharp decline in visitors (affecting total visitor expenditure) and the reactionary increase in federal spending.

Limitations and Future Research: 1. While this time-series analysis provides robust insights, it had violated the random sampling assumption. However, OLS estimators remain unbiased provided the series is weakly dependent.

2. Furthermore, I identified that Defense Spending per Personnel is a relevant instrument ($p < 0.05$) for Federal Spending. My future research should utilize this instrument to isolate and find the true coefficient of non-defense spending. Additionally, I should look to investigate the “U-shape” observed in tourism, which may reflect the economic costs of over-tourism that I have omitted like infrastructure limits at high visitor levels.